L1

CLASS X - SCIENCE

HUMAN EYE AND THE COLOURFUL WORLD

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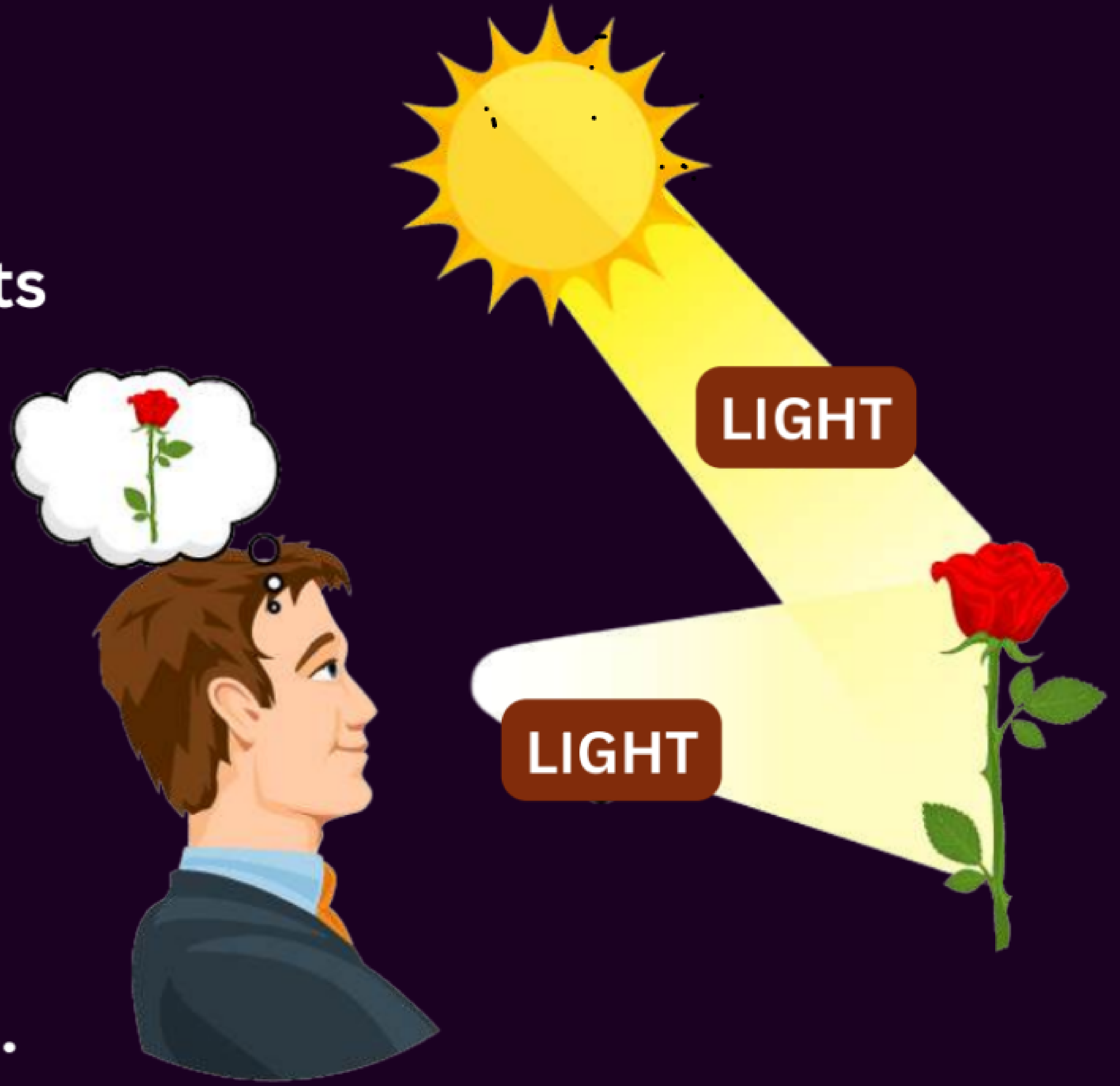
आरम्भ
BATCH
JOSH
METER
LOW
MODERATE
HIGH

Why Do We See Objects?

- We cannot see objects in a dark room.
- When light falls on an object, the object reflects light.
- This reflected light enters our eyes.
- Hence, we are able to see objects.



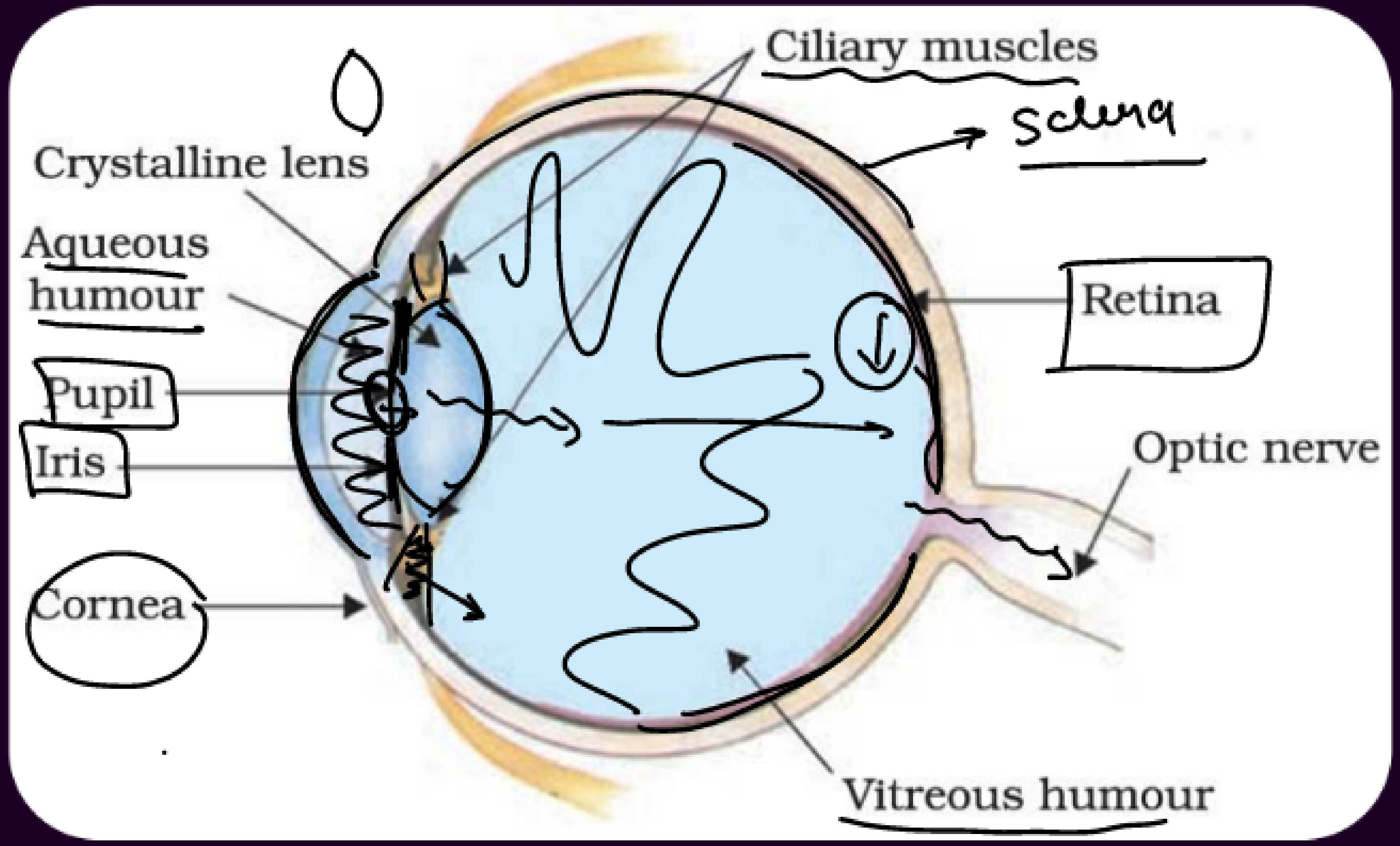
Ray of Light: The straight-line path of light is represented by a ray of light.

- A ray shows the direction in which light travels.

Human Eye and it parts

1. Cornea:

- Light enters the eye through a thin transparent membrane called the cornea.
- ***It forms a transparent bulge on the front surface of the eyeball.***
- Most of the refraction of light entering the eye occurs at the outer surface of the cornea.

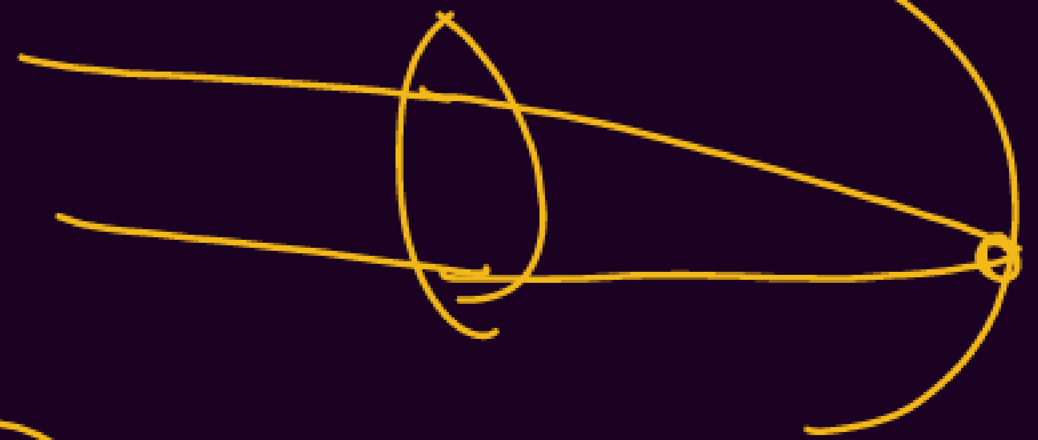


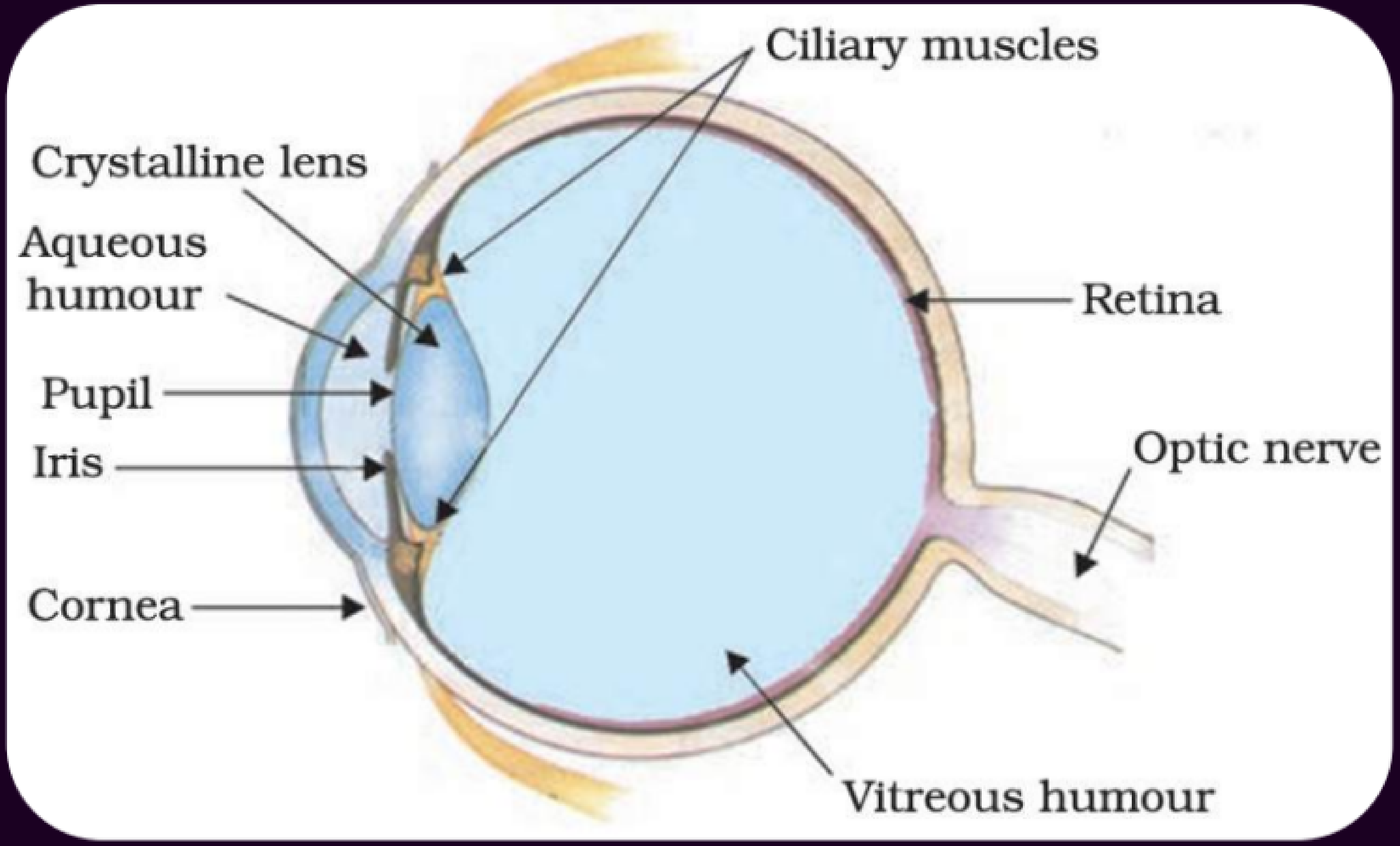
2. Eyeball:

- The eyeball is approximately spherical in shape.
- Its **diameter is about 2.3 cm**.

3. Eye Lens / Crystalline Lens

- The eye lens is a transparent crystalline lens.
- It helps in fine adjustment of **focal length**.
- It focuses objects at different distances on the retina.





4. Iris:

- Iris is a **dark muscular diaphragm** present behind the cornea.
- It controls the size of the pupil.

More light → small
less light → Big

5. Pupil

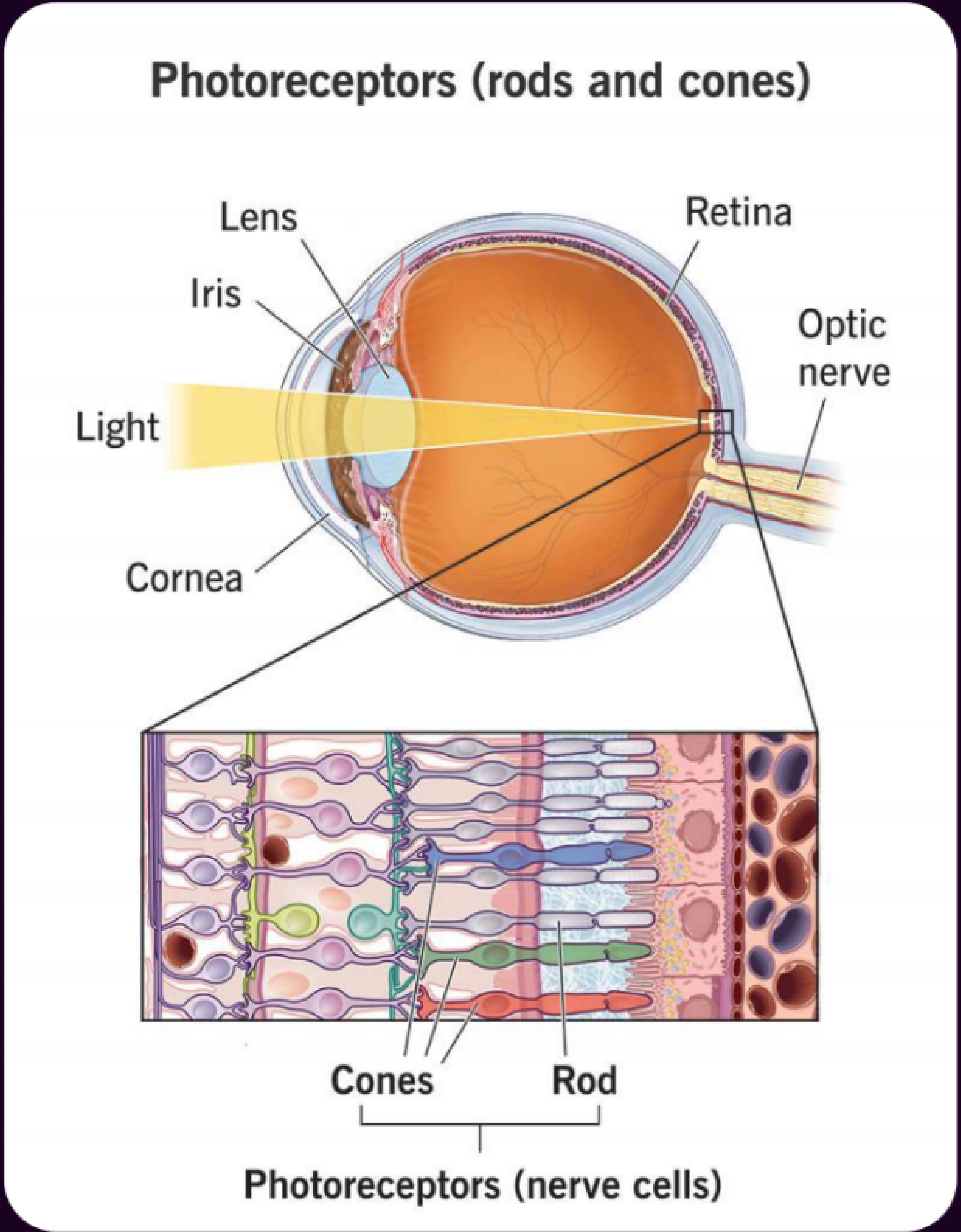
- Pupil is the **opening in the centre of the iris**.
- It controls and regulates the amount of light entering the eye.

6. Retina

- Retina is a delicate, light-sensitive membrane at the back of the eye.
- The eye lens forms a **real and inverted** image on the retina.
- Retina contains a large number of light-sensitive cells called **photoreceptors**.

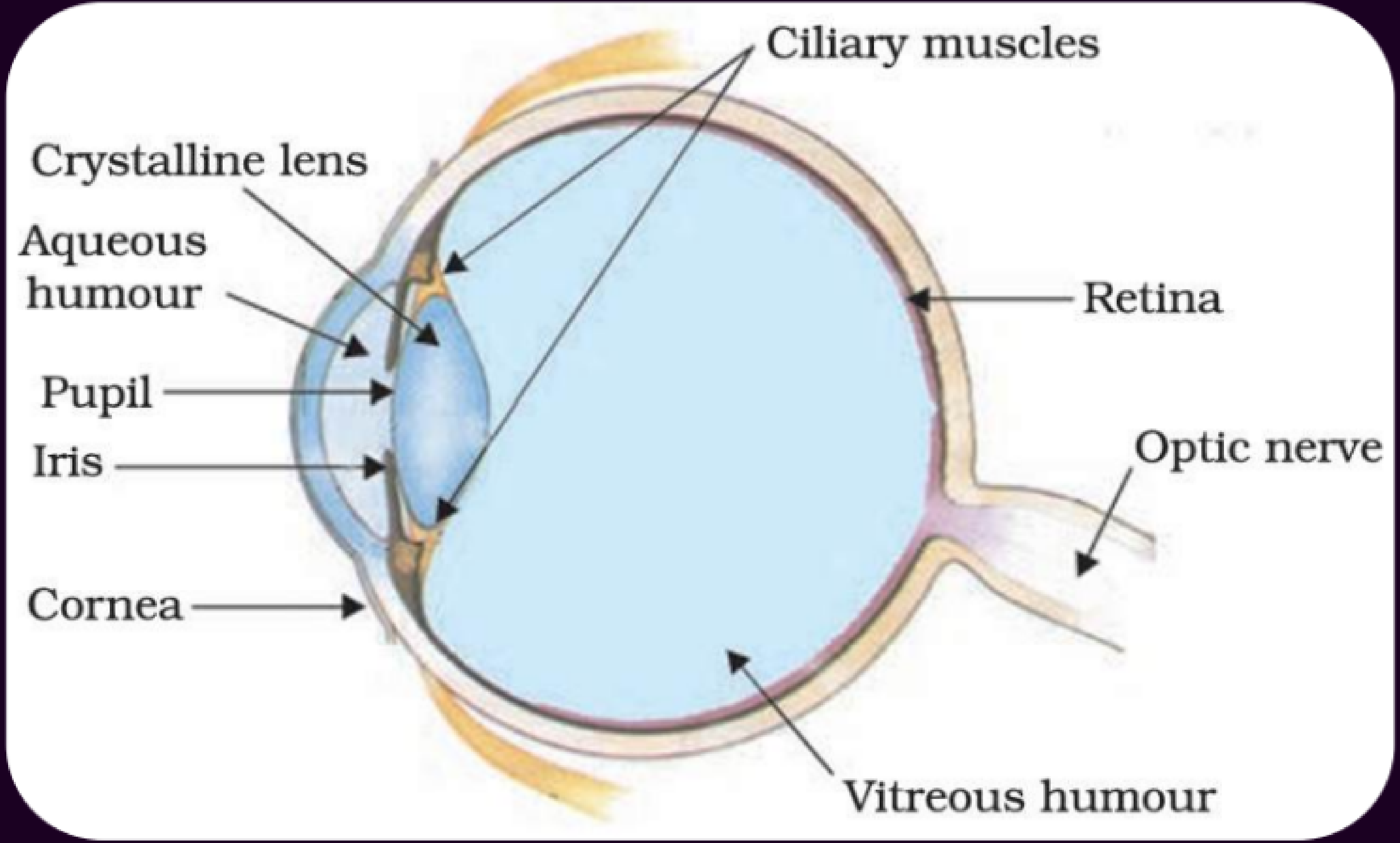
These photoreceptors are of two types: ***rods and cones***.

1. **Rods** help in dim light and black-and-white vision.
2. **Cones** help in bright light and colour vision.



7. Optic Nerve

- Light-sensitive cells of the retina generate **electrical signals.**
- These signals are sent to the brain through the optic nerve.
- The brain interprets these signals and helps us see objects as they are.



8. Aqueous Humour

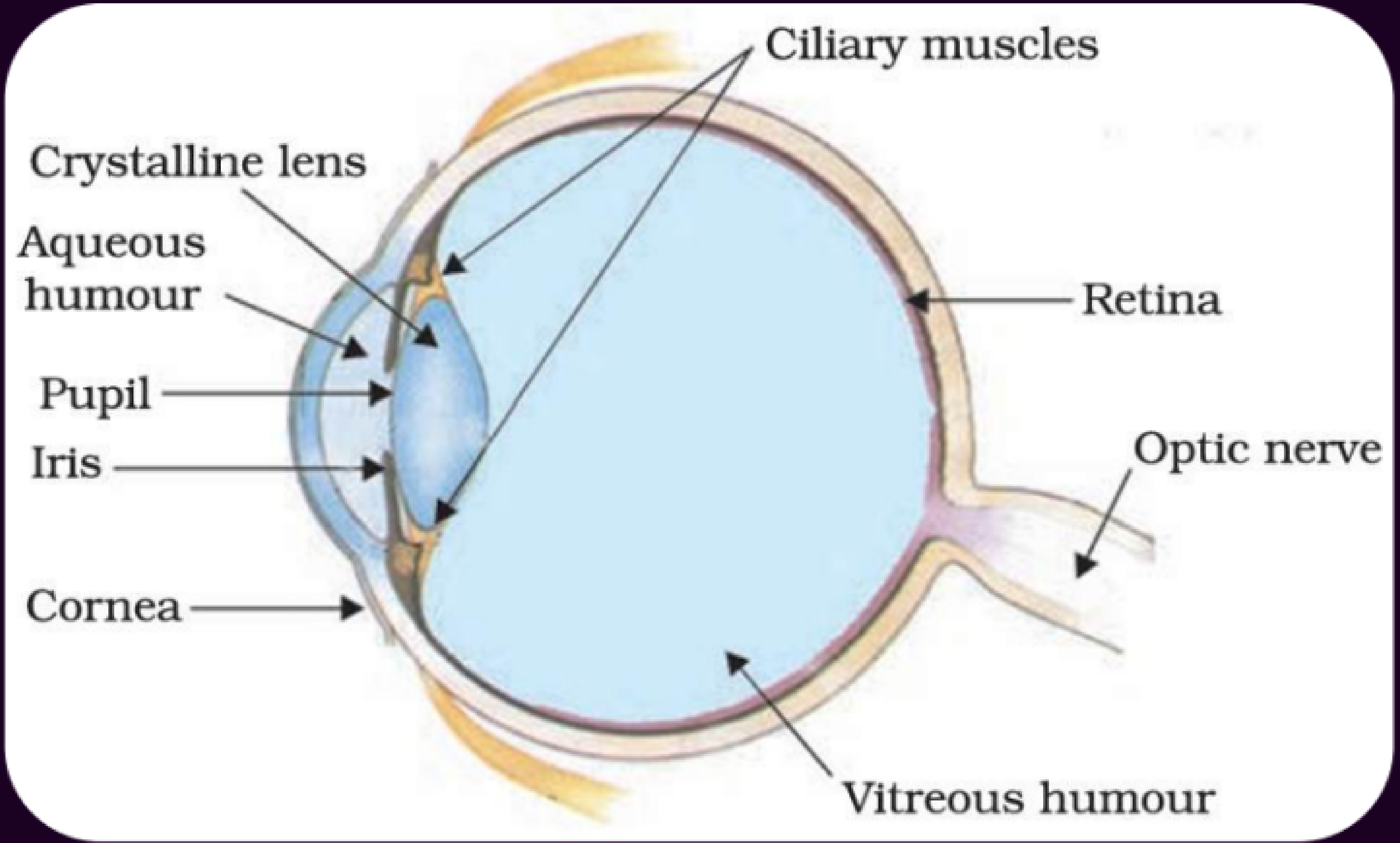
- Aqueous humour is a transparent **watery fluid** present between the cornea and the eye lens.
- It nourishes the cornea and the lens.
- It **removes waste materials** from this region of the eye.
- It helps to maintain the shape of the eye.
- It also helps to maintain the internal pressure of the eye.

9. **Ciliary muscles –**

Change the shape of the lens for near and distant vision.

10. **Vitreous humour –**

Jelly-like substance that maintains the shape of the eye.



Advantage of eyes in front of the face:

- Helps us see objects clearly
- It gives a wider field of view.
- It provides three-dimensional view
- Both eyes work together for better vision

Soch ke batao

Q. The part of the human eye which controls the amount of light entering into it: [CSBE 2025]

(a) Iris
(b) Cornea
(c) Ciliary muscles
(d) Pupil

Soch ke batao

Q. The lens system of human eye forms an image on a light sensitive screen, which is called as: [CBSE 2024]

(a) Cornea
(b) Ciliary muscles
(c) Optic nerves
(d) Retina

Soch ke batao

Q. When you look at an object very close to your eyes, the:

(A) Ciliary muscles of your eye contract and the eye lens becomes thick.

(B) Ciliary muscles of your eye get relaxed and the eye lens becomes thick.

(C) Ciliary muscles of your eye contract and the eye lens becomes thin.

(D) Ciliary muscles of your eye get relaxed and the eye lens becomes thin.

Soch ke batao

Q. What is the function of ciliary muscles? [2012, 2015, 2018]

Answer:

Control the curvature of the crystalline lens and change its power for clear vision.

Soch ke batao

Q. Write the function of each of the following parts of the human eye: [CBSE 2018]

(i) Cornea

(ii) Iris

(iii) Crystalline lens

(iv) Ciliary muscles

(i) Cornea – The transparent outer layer of the eye that helps in refraction and focusing of light entering the eye.

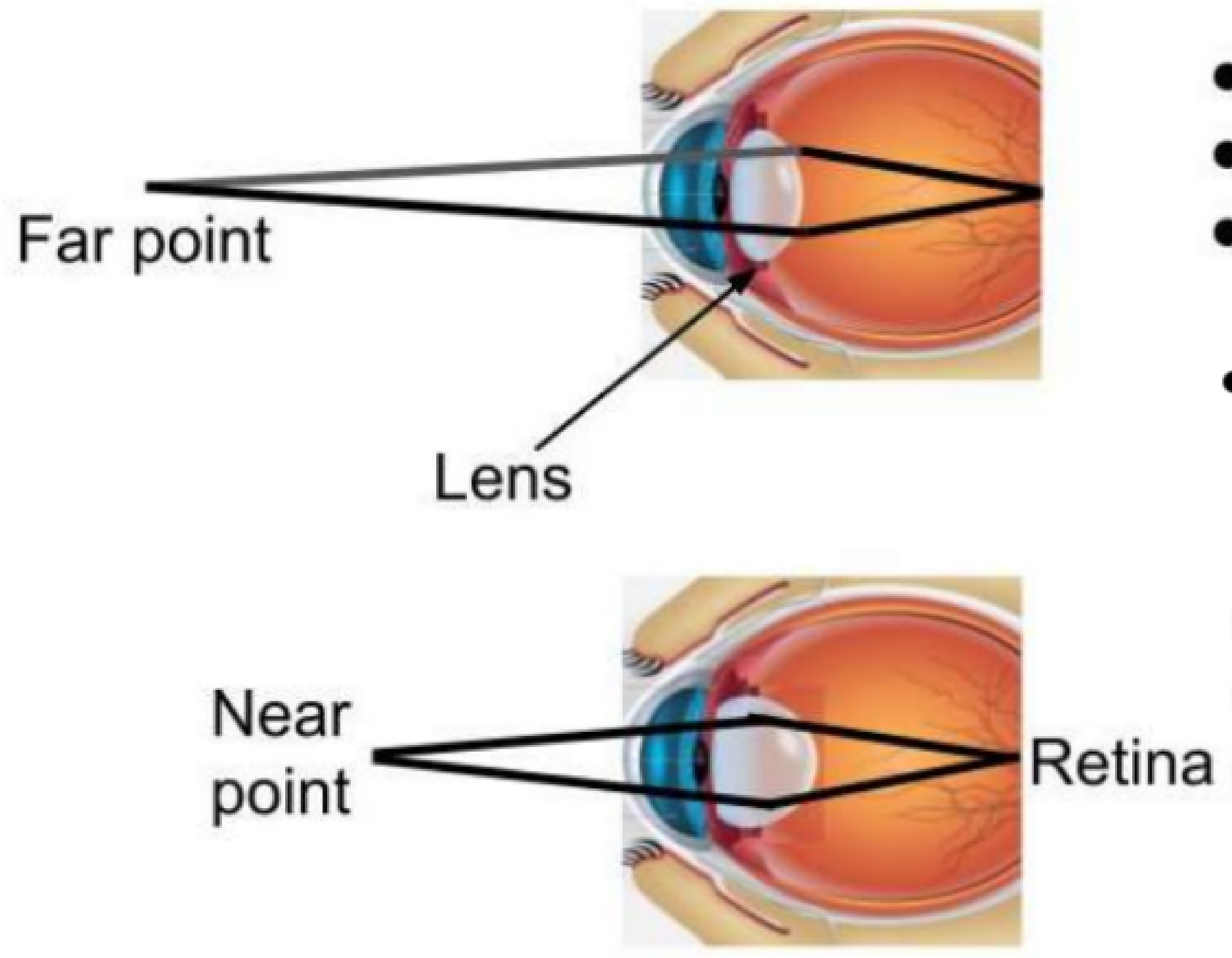
(ii) Iris – Controls the size of the pupil and the amount of light entering the eye.

(iii) Crystalline lens – Adjusts the focal length to focus objects at different distances on the retina.

(iv) Ciliary muscles – Control the curvature of the crystalline lens and change its power for clear vision.

Power of Accommodation

- The eye lens is flexible and its shape can be changed by the ciliary muscles.
- Changing the shape of the lens changes its **focal length.**
- The ability of the eye lens to adjust its focal length to clearly focus both near and distant objects on the retina is called the **power of accommodation** of the eye.



- Far Object
- Expanded Eye lens
- Focal length increases and power decreases
- **Ciliary muscles relax → lens becomes thin → focal length increases**

- Nearby Object
- Contracted Eye lens
- Focal length decreases and power increases
- **Ciliary muscles contract → lens becomes thicker → focal length decreases**